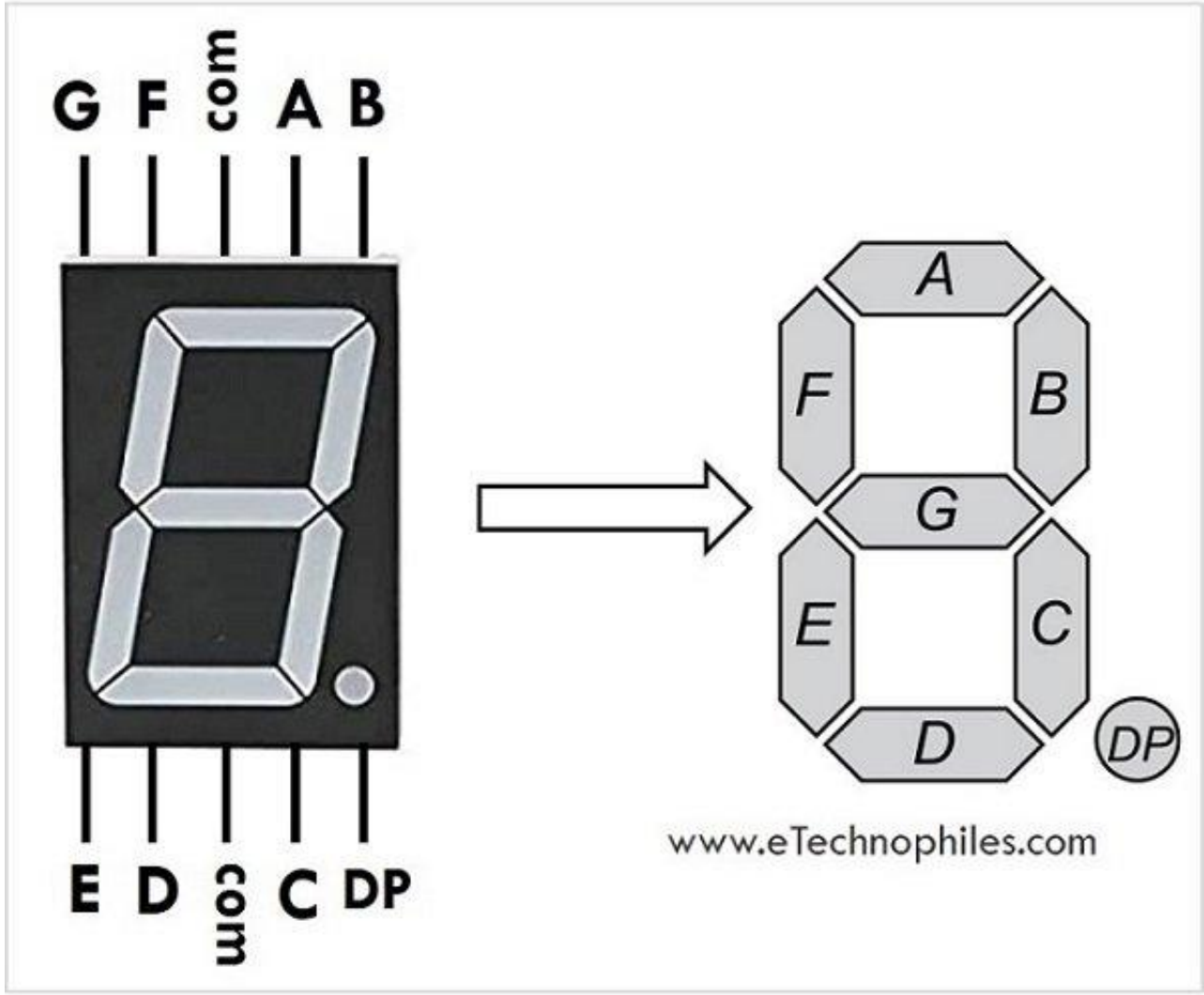
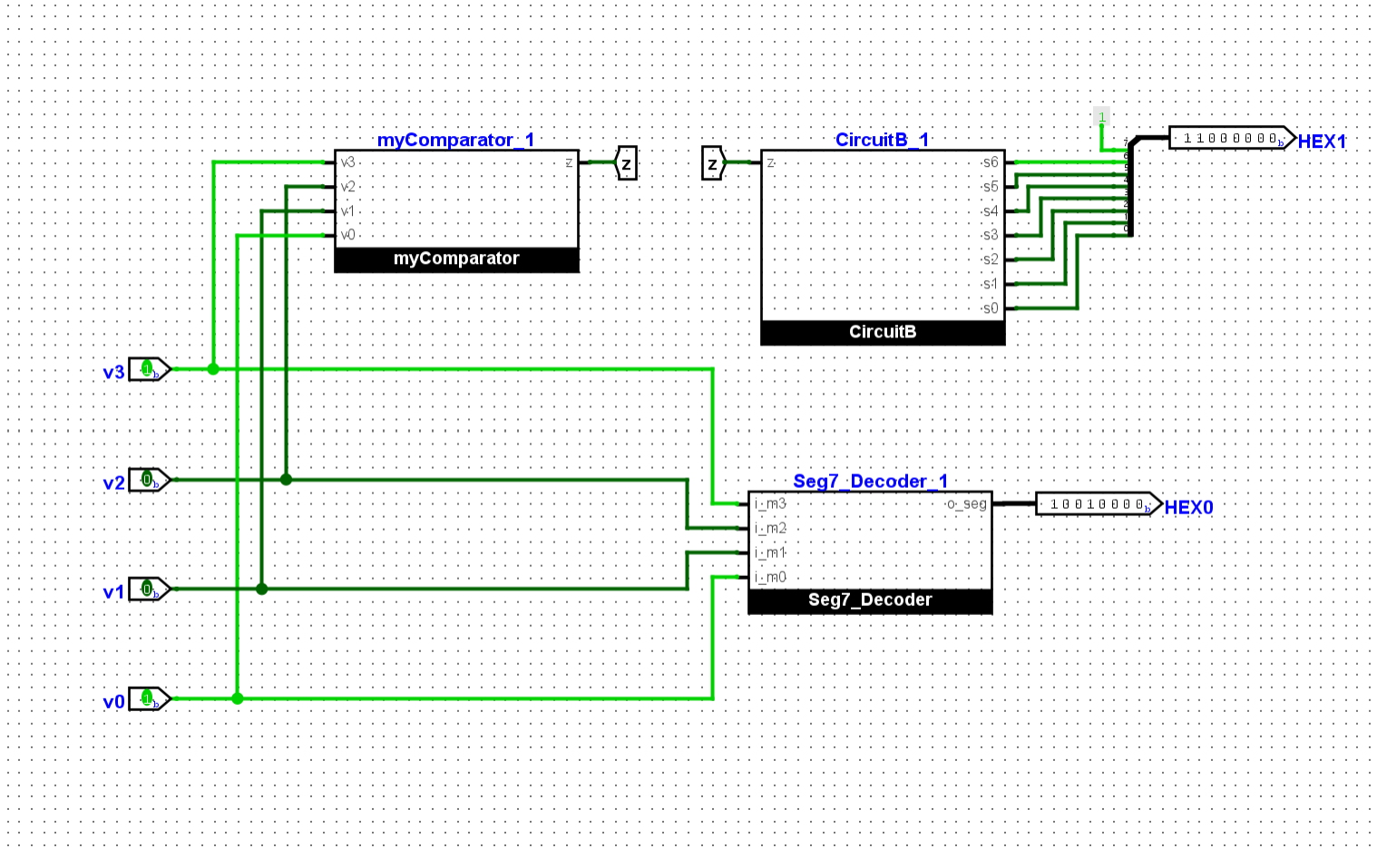
Lab 4

By Gabby Vargas Leon

7 segment display layout



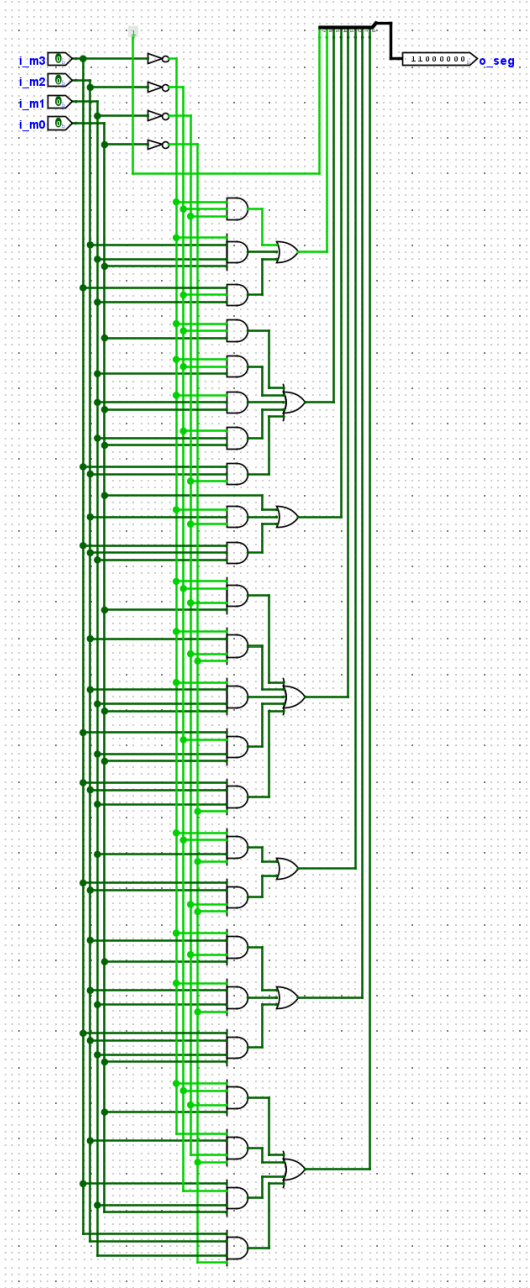
Bin to Hex circuit and truth table



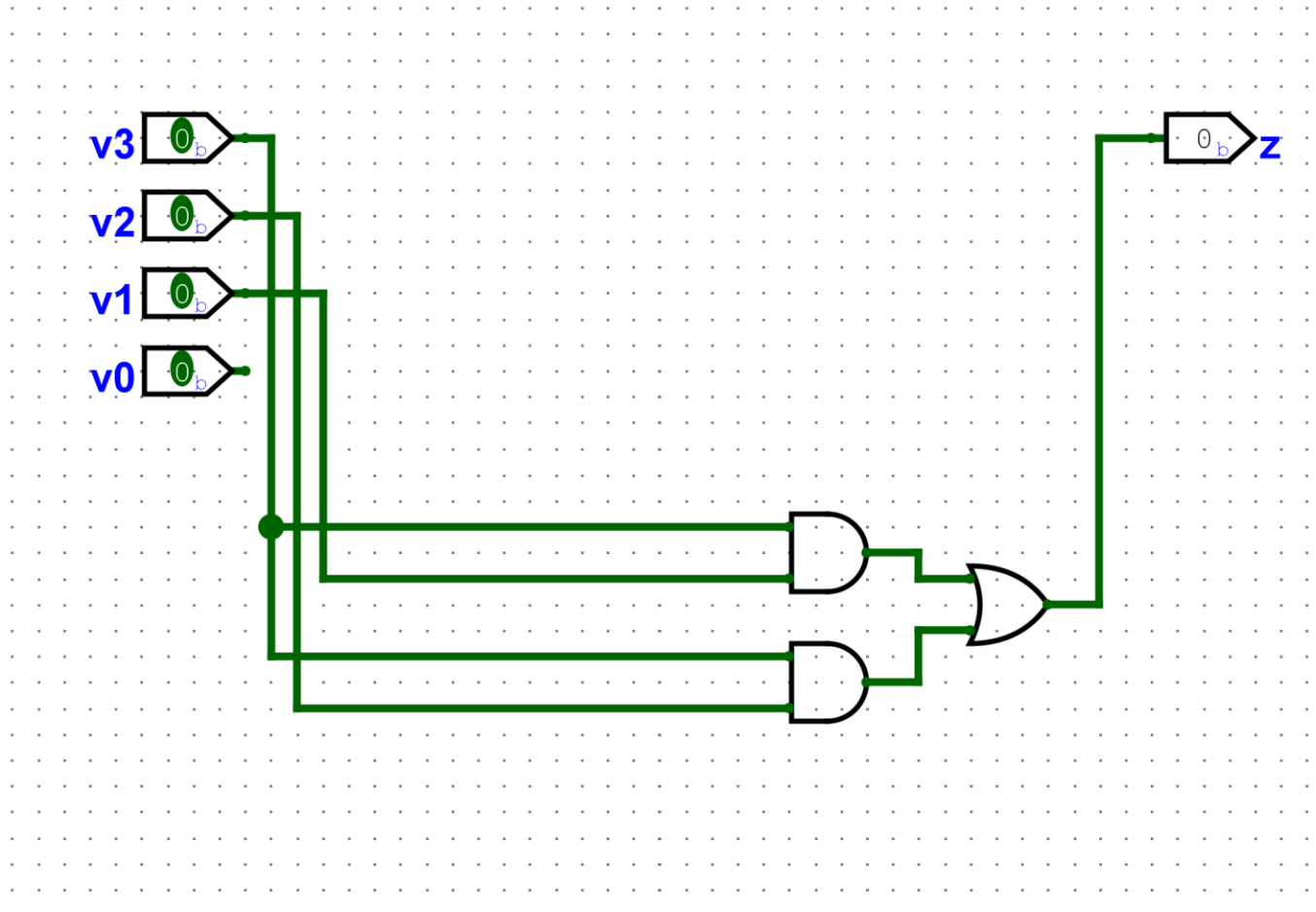
v3	v2	v1	v0	HEX1[7..0]	HEX0[7..0]
0	0	0	0	1 1 0 0 0 0 0 0	1 1 0 0 0 0 0 0
0	0	0	1	1 1 0 0 0 0 0 0	1 1 1 1 1 0 0 1
0	0	1	0	1 1 0 0 0 0 0 0	1 0 1 0 0 1 0 0
0	0	1	1	1 1 0 0 0 0 0 0	1 0 1 1 0 0 0 0
0	1	0	0	1 1 0 0 0 0 0 0	1 0 0 1 1 0 0 1
0	1	0	1	1 1 0 0 0 0 0 0	1 0 0 1 0 0 1 0
0	1	1	0	1 1 0 0 0 0 0 0	1 0 0 0 0 0 1 0
0	1	1	1	1 1 0 0 0 0 0 0	1 1 1 1 1 0 0 0
1	0	0	0	1 1 0 0 0 0 0 0	1 0 0 0 0 0 0 0
1	0	0	1	1 1 0 0 0 0 0 0	1 0 0 1 0 0 0 0
1	0	1	0	1 1 1 1 1 0 0 1	1 1 0 0 0 0 0 0
1	0	1	1	1 1 1 1 1 0 0 1	1 1 1 1 1 0 0 1
1	1	0	0	1 1 1 1 1 0 0 1	1 0 1 0 0 1 0 0
1	1	0	1	1 1 1 1 1 0 0 1	1 0 1 1 0 0 0 0
1	1	1	0	1 1 1 1 1 0 0 1	1 0 0 1 1 0 0 1
1	1	1	1	1 1 1 1 1 0 0 1	1 0 0 1 0 0 1 0

7 segment display truth table showing 0-15 and circuit that was build by the truth table

i_m3	i_m2	i_m1	i_m0	o_seg[7..0]
0	0	0	0	1 1 0 0 0 0 0 0
0	0	0	1	1 1 1 1 1 0 0 1
0	0	1	0	1 0 1 0 0 1 0 0
0	0	1	1	1 0 1 1 0 0 0 0
0	1	0	0	1 0 0 1 1 0 0 1
0	1	0	1	1 0 0 1 0 0 1 0
0	1	1	0	1 0 0 0 0 0 1 0
0	1	1	1	1 1 1 1 1 0 0 0
1	0	0	0	1 0 0 0 0 0 0 0
1	0	0	1	1 0 0 1 0 0 0 0
1	0	1	0	1 1 0 0 0 0 0 0
1	0	1	1	1 1 1 1 1 0 0 1
1	1	0	0	1 0 1 0 0 1 0 0
1	1	0	1	1 0 1 1 0 0 0 0
1	1	1	0	1 0 0 1 1 0 0 1
1	1	1	1	1 0 0 1 0 0 1 0

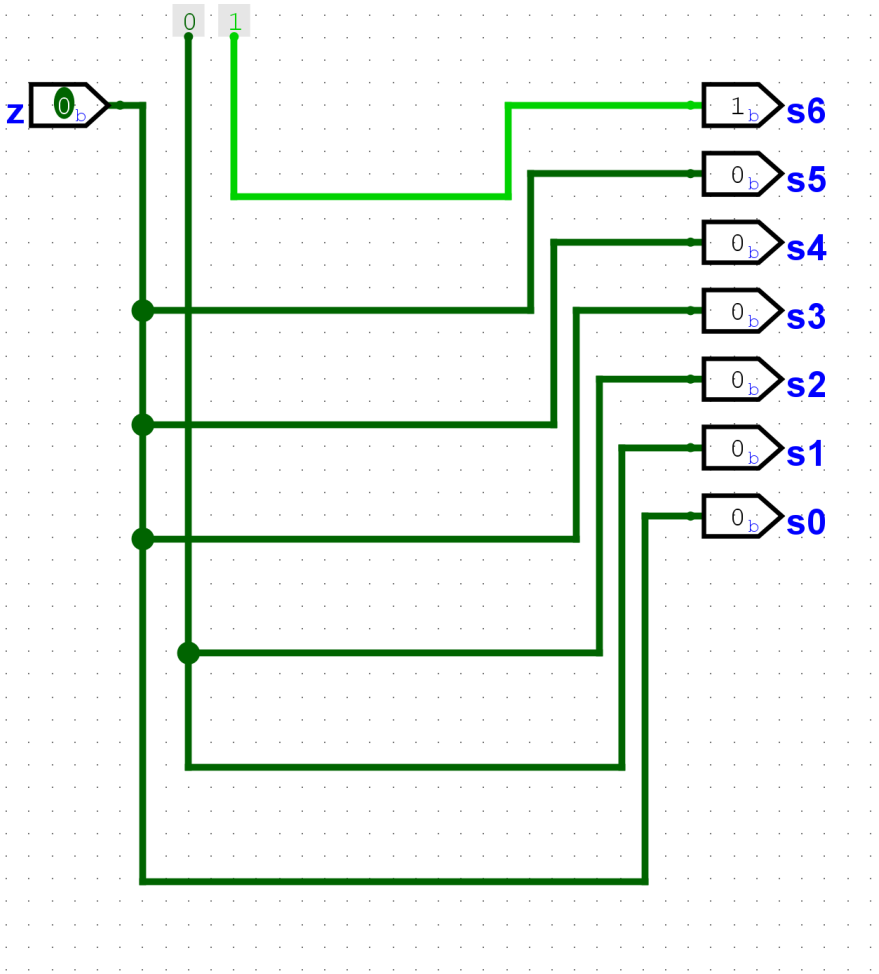


Comparator circuit built by the truth table



v3	v2	v1	v0	z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Carry over bit when the first 7-segment display goes over 9



2 of 2 rows shown

z	s6	s5	s4	s3	s2	s1	s0
0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	1

Finale circuit look

